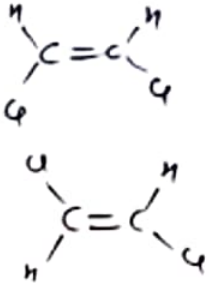
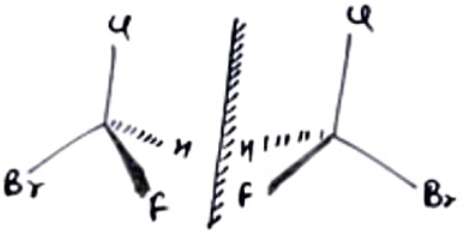
CONFORMATIONAL ISOMERISM

Gr I



Optical isomers



CONFORMATIONAL ISOMERISM

Definitions

① Conformations:- Different 3D arrangement of atoms or groups of a compound which are interconvertible into each orther at room temp by rotation of σ bonds are called Conformations.

Stable conformations are called conformer = Rotamer = Conformational iso.

CONFORMATIONAL ISOMERISM

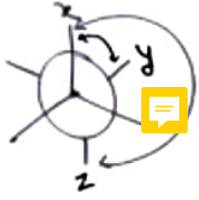
Definitions

① **Conformations:-** we can not separate these isomers

② **Dihedral angle / Torsional angle :-**

Angle between two bonds which are present of two adjacent atoms.

(angle of c-x and c-y in x-c-c-y)



CONFORMATIONAL ISOMERISM

Definitions

③ Staggered conformations:-

Conformations in which value of dihedral angle = 60°, 180°, 300°



④ Eclipsed conformation:-

Conformation in which value of dihedral angle = 0°, 120°, 240°



⑤ Skew conformation:-

Conformations other than staggered and eclipsed.

CONFORMATIONAL ISOMERISM

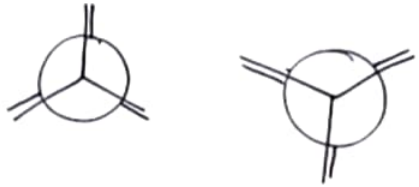
Representation of Staggered and Eclipsed

Staggered

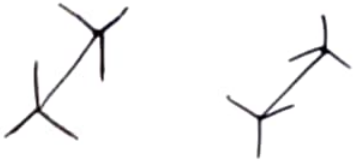
Eclipsed

⇒ Newmann





⇒ Sawhorse





⇒ Wedge-dash





or

CONFORMATIONAL ISOMERISM

⑥ Energy barrier = Conformational energy:-

Energy required for complete rotation of a bond.

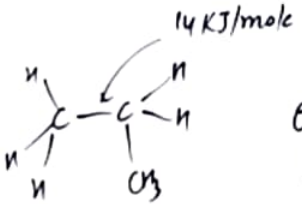
or

Energy difference between most and least stable conformation.

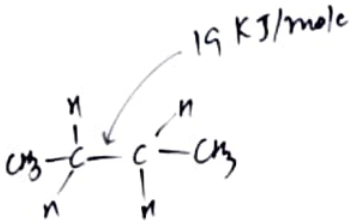
eg 1



eg 2



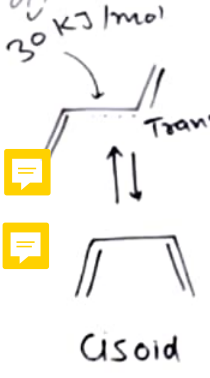
eg 3



CONFORMATIONAL ISOMERISM

⑥ Energy barrier = Conformational energy :-

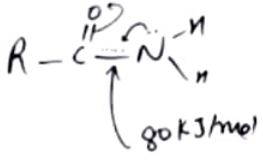
eg 4



eg 5



eg 6



⇒ Energy barrier ∝ presence of bulky groups

⇒ ∴ ∴ ∴ ∝ multiple bond chacter

CONFORMATIONAL ISOMERISM

⑦ Factors affecting stability of conformations:-

Ⓐ Torsional Strain = Eclipsed Strain:-



Repulsion between electron density of two dihedral bonds.

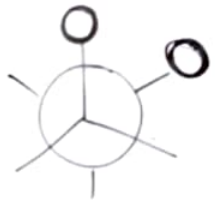
⇒ Eclipsed > Staggered (Torsional Strain)

⇒ Staggered > Eclipsed (Stability of conformation)

CONFORMATIONAL ISOMERISM

⑦ Factors affecting stability of conformations:-

ⓑ Vander Waal's Strain:-



Repulsion between groups which are present on dihedral bonds.

$-O-H^{\delta+}$:F –
$-N-H^{\delta+}$:O<
$F-H^{\delta+}$:N<

ⓒ Intramolecular hydrogen bonding.

Staggered conformations which have intramolecular hydrogen bonding is/are most stable.

CONFORMATIONAL ISOMERISM

⑦ Factors affecting stability of conformations:-

Ⓓ Gauche effect

Ⓔ Anomeric effect

Ⓕ Ring strain.

Factors affecting stability of conformers:
. Torsional strain
. Vander Waal strain
. Intramolecular Hydrogen bonding
. Gauche effect
. Anomeric effect
. Ring strain effect

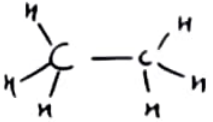
Ⓒ Intramolecular hydrogen bonding.

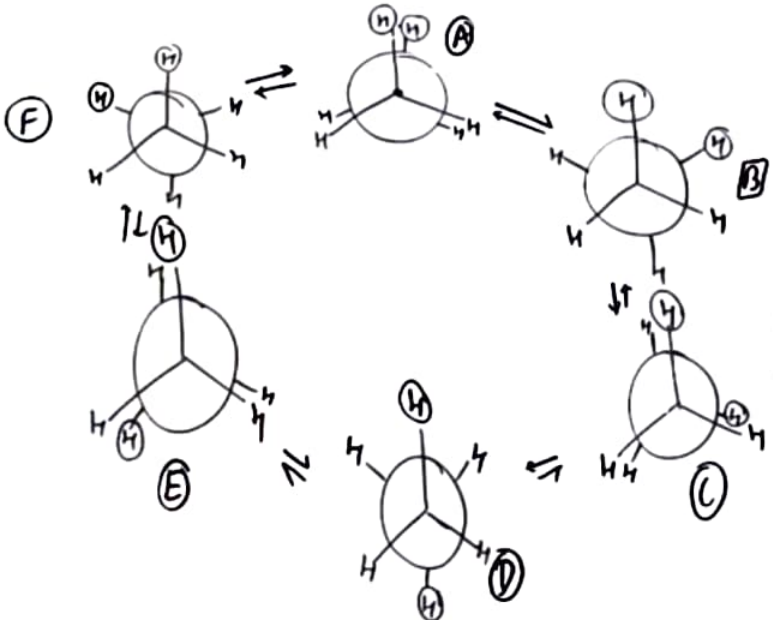
Staggered conformations which have intramolecular hydrogen bonding is/are most stable.

CONFORMATIONAL ISOMERISM

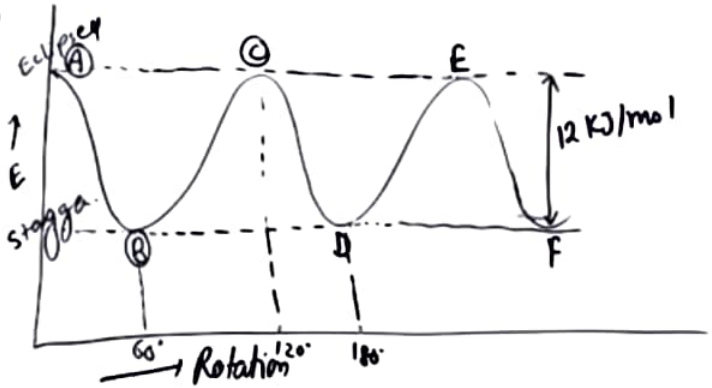
Examples

① Ethane



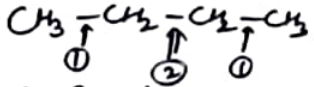


- Stability ~100% Staggered > Eclipsed ~0%
- no of conformations = ∞
- no of conformational isomers = 3
- energy barrier = 12 KJ/mol

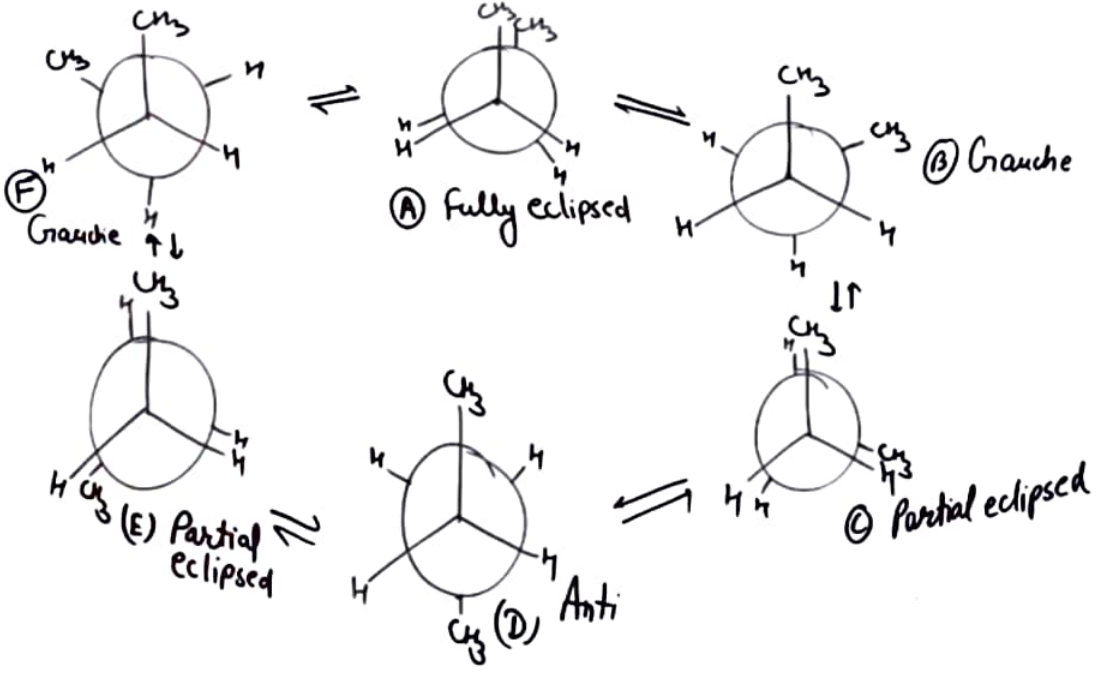


CONFORMATIONAL ISOMERISM

eg:- Butane

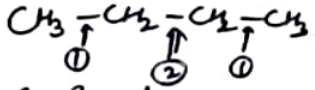


⇒ W.r.t. C_2-C_3 Bond.



CONFORMATIONAL ISOMERISM

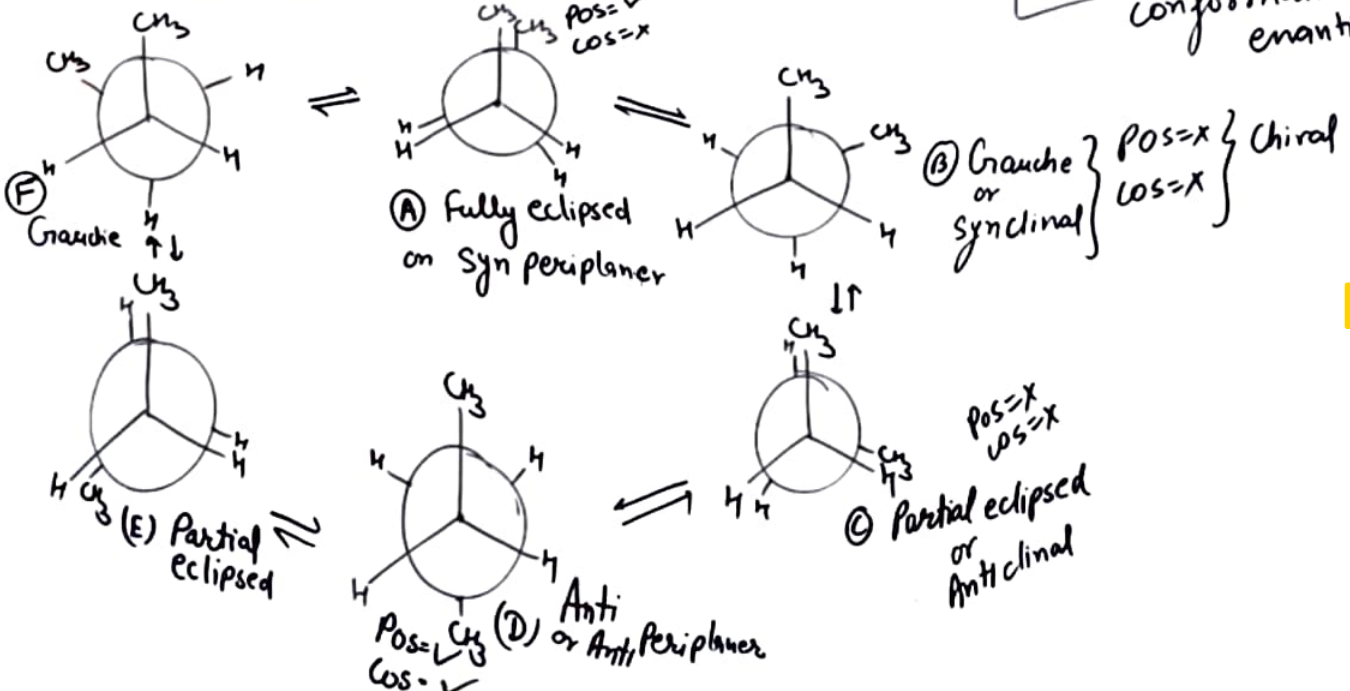
eg:- Butane



→ W.r.t. C_2-C_3 Bond.

→ 

conformational enantiomers.



CONFORMATIONAL ISOMERISM

⇒ Stability of conformation of butane :-

Anti > Gaucche >>> Partial eclipsed > fully eclipsed

~100% ~0%

⇒ no. of conformation = ∞

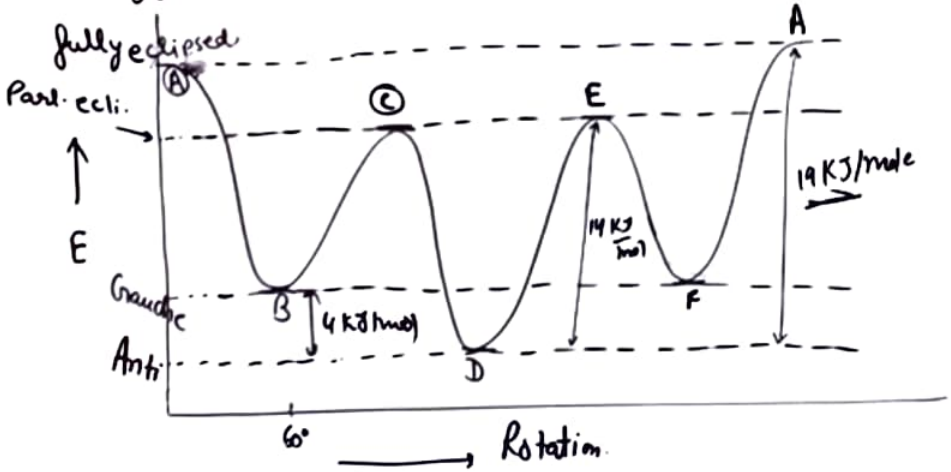
⇒ no. of conformational isomers = 3

CONFORMATIONAL ISOMERISM

- Stability order of conformations of butane :-

Anti > Gauche >> Partial eclipsed > fully eclipsed

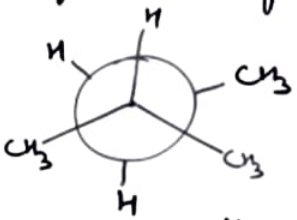
- Energy graph for conformations of butane



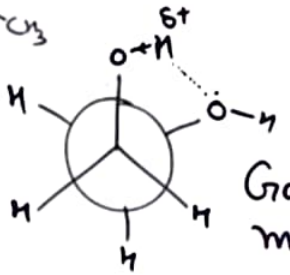
CONFORMATIONAL ISOMERISM

Q Draw most Stable conformation of given compounds.

① $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_3$ ⇒



② $HO-CH_2-CH_2-OH$ ⇒



Gauche most Stable

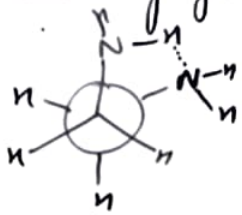
③ $H-O-CH_2-CH_2-F$
④ $H-O-CH_2-CH_2-CHO$
⑤ $H-O-CH_2-CH_2-COOH$
⑥ $H-O-CH_2-CH_2-\ddot{N}H_2$

Gauche > Staggered (Stability)

CONFORMATIONAL ISOMERISM

Q Draw most Stable conformation of given compounds.

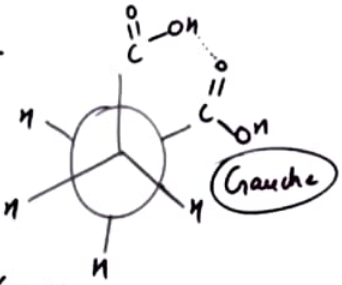
⑦ $\ddot{N}H_2-CH_2-CH_2-\ddot{N}H_2$



Gauche

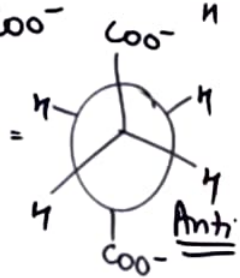
⑧ Succinic acid

in acidic med. → COOH, COOH



Gauche

basic medium → COO^-, COO^-



Anti

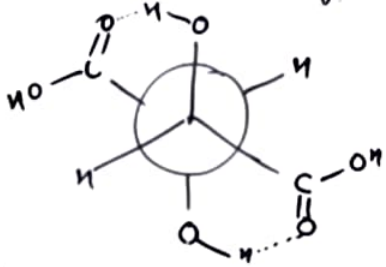
CONFORMATIONAL ISOMERISM

Q Draw most Stable conformation of given compounds.

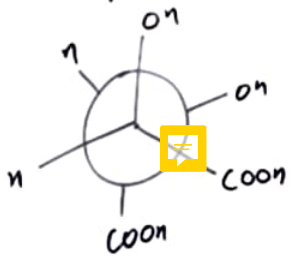
eg 9



Meso Tartaric acid



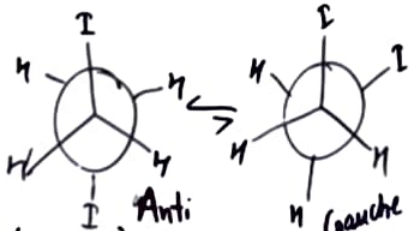
Anti
most Stable



Gauche form
lesser stable

eg 10 $I-CH_2-CH_2-I$

$\mu_{Net} \neq 0$



Anti
most Stable

$\mu_A = 0$

X_A

Gauche

$\mu_G \neq 0$

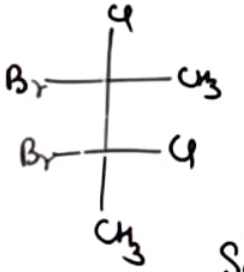
X_G

$\mu_{Net} = \mu_A X_A + \mu_G X_G$

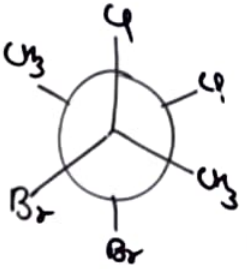
$\mu_A = 0$

CONFORMATIONAL ISOMERISM

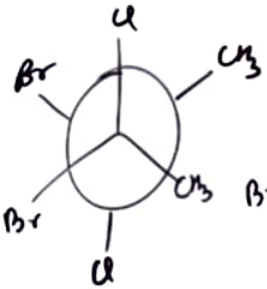
Q The total number(s) of Stable conformers with non-zero dipole moment for the following compound is/are. [JEE. ADV. 2014]



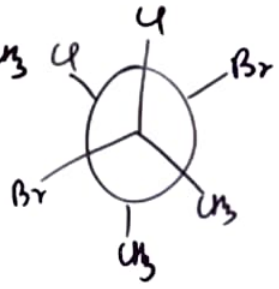
Sol.



$\mu \neq 0$



$\mu \neq 0$



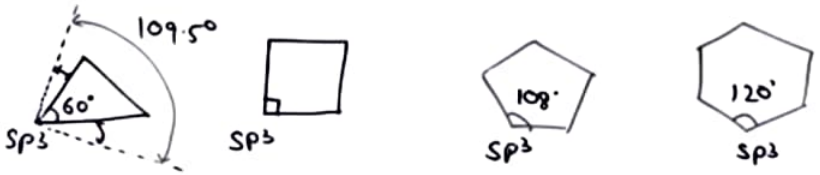
$\mu \neq 0$

Ans 3

CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

Baeyer's Ring Strain:- It is represented by difference between actual bond angle and ideal bond angle in rings.



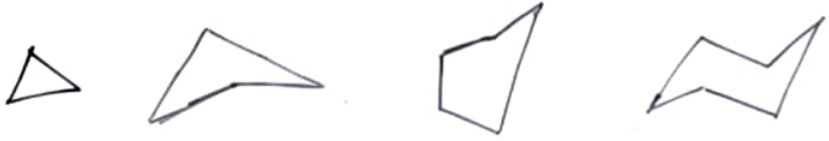
ideal angle =	109.5°	109.5°	109.5°	109.5°
actual ...	60°	90	108	120
(if planar) ring	49.5°	19.5°	1.5°	10.5°

CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

5 > 6 > 4 > 3 stability (incorrect)

Stability (exp.) ⇒ cyclohexane > cyclopentane > cyclobutane > cyclopropane



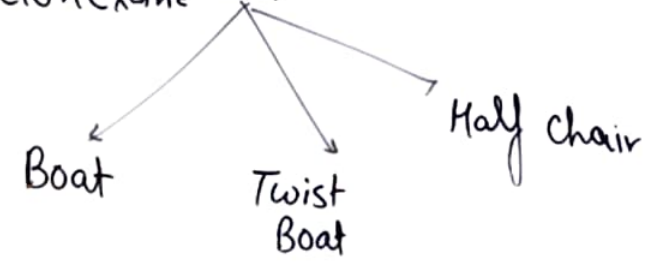
CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

- Cyclohexane is non planar structure
- It is exist in chair conformation.

⇒ The chair conformations are interconvertible into each other.

⇒ Other conformations of cyclohexane are

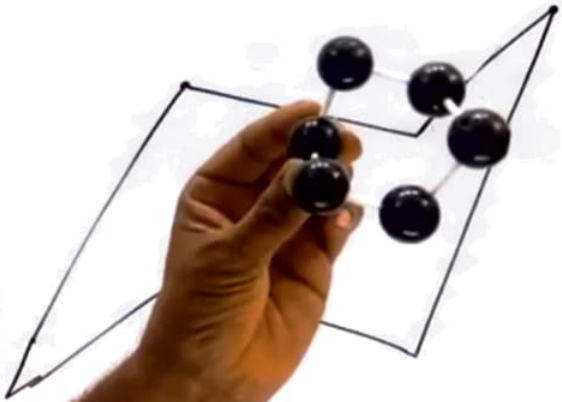


ONFORMATIONAL ISOMERISM

F CYCLOHEXANES

fo. of cyclohexane:

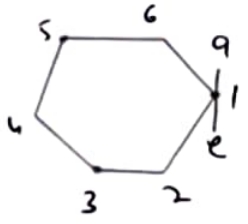


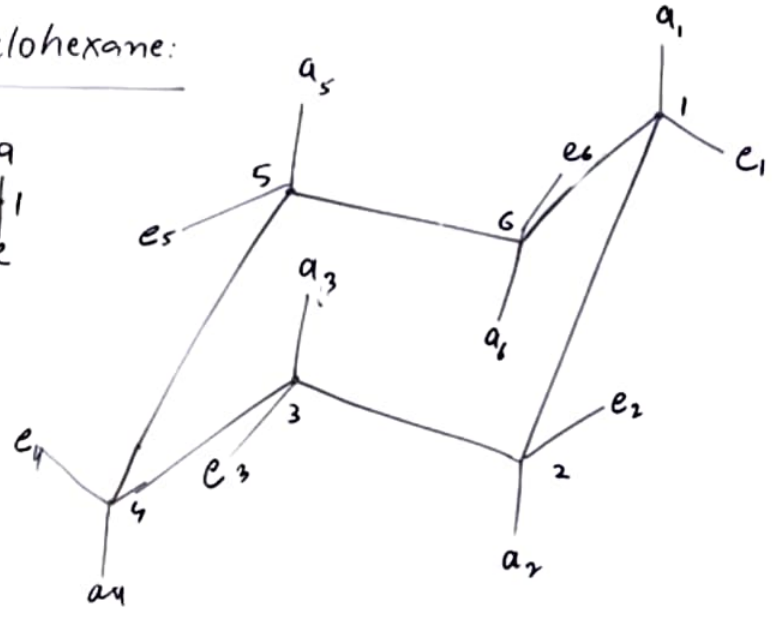


CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

(A) Chair conf. of cyclohexane:





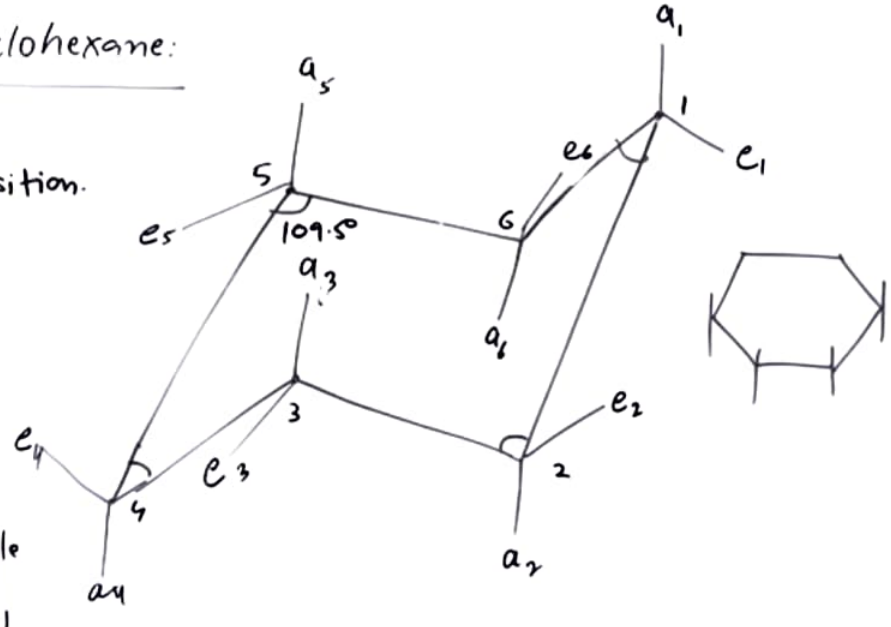
axial = a

e = equitorial

CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

(A) Chair conf. of cyclohexane:



⇒ 6 axial & 6 - equitorial position.

⇒ $C_1-C_2 \,||\, C_4-C_5$

$C_2-C_3 \,||\, C_5-C_6$

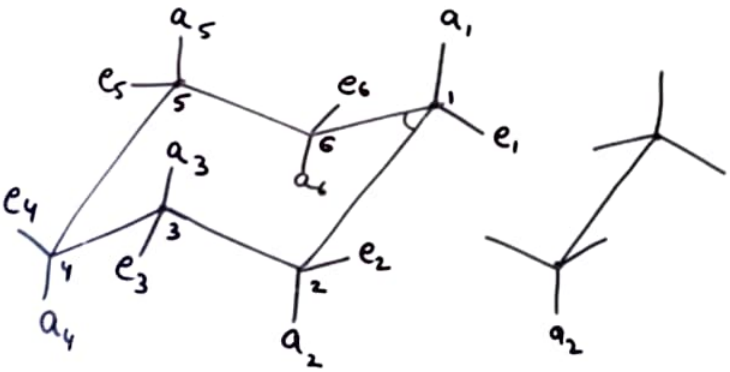
$C_3-C_4 \,||\, C_1-C_6$

⇒ No angle strain (bond angle = 109.5°)

CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

(A) Chair conformation of cyclohexane

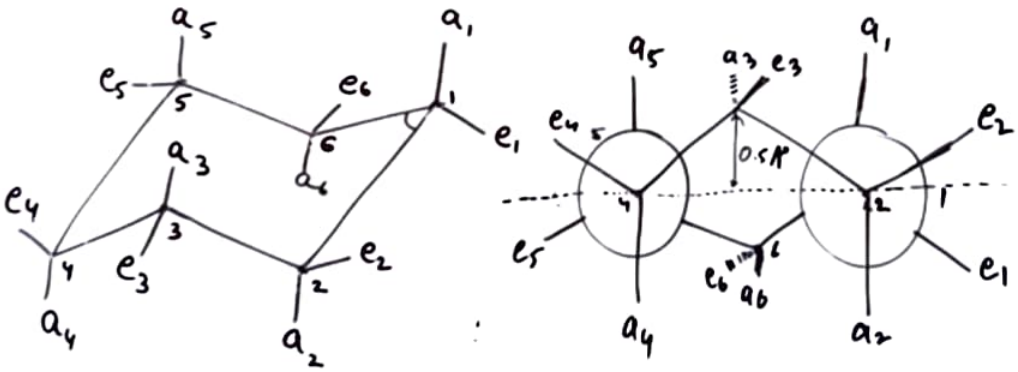


- No angle Strain
  (Bond angle = 109.5°)
- No Torsional Strain
  (Staggered conformation)

CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

(A) Chair conformation of cyclohexane

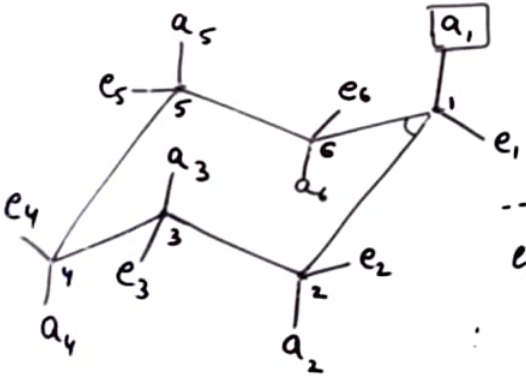


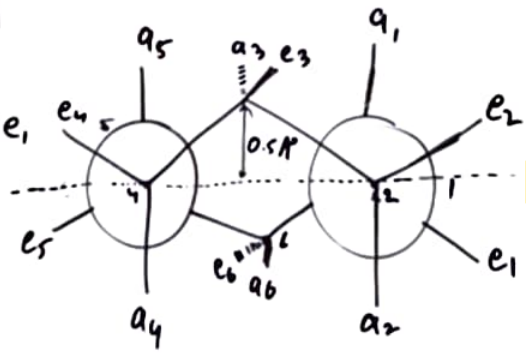
- No angle Strain (Bond angle = 109.5°)
- No Torsional Strain (Staggered conformation)

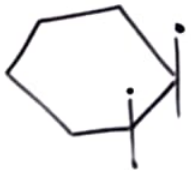
CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

(A) Chair conformation of cyclohexane







- No angle Strain (Bond angle = 109.5°)
- No Torsional Strain (Staggered conformation)
- identification of Cis-Trans.

① Cis position w.r.t a_1 = e_2, a_3, e_4, a_5, e_6

② Trans " " " $a_1 = a_2, e_3, a_4, e_5, a_6$

③ Cis " " $e_1 = a_2, e_3, a_4, e_5, a_6$

④ Trans " " " $e_1 = e_2, a_3, e_4, a_5, e_6$

CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

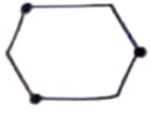
(A) Chair conformation of cyclohexane

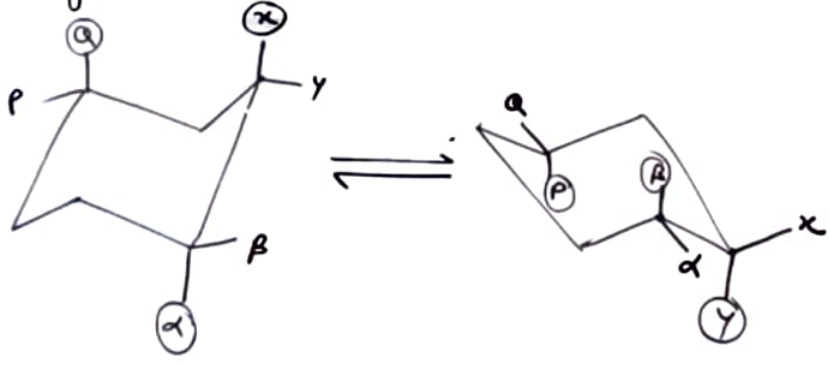
- These chair conformations are interconvertible which is known flippling in cyclohexane and other conformations are form between two chair conformations.
- During flippling in cyclohexane (chair conformation) axial and equitorial position are interchange. (axial ⇄ equitorial)

CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

(A) Chair conformation of cyclohexane



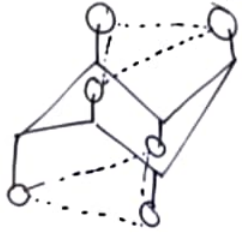


CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

(A) Chair conformation of cyclohexane

- alternate axial positions are more closer than equitorial (a_1, a_3, a_5) & (a_2, a_4, a_6) so these axial positions are more steric hindered position therefor bulky group is more Stable on equitorial position.

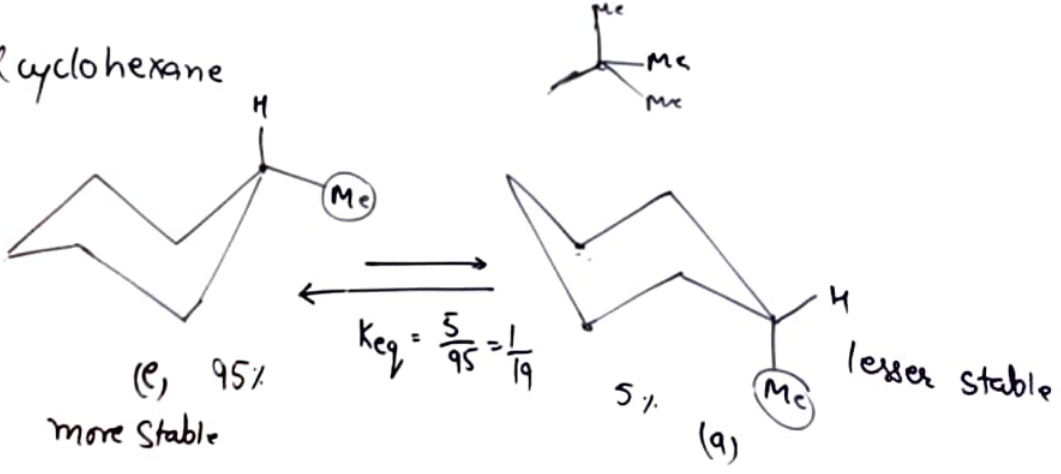


CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

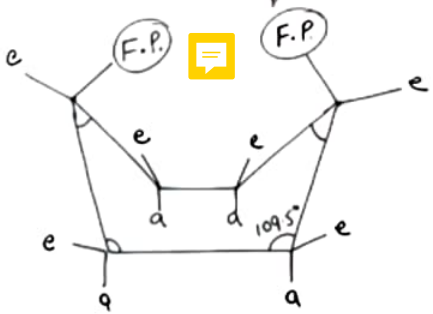
(A) Chair conformation of cyclohexane

eg. Methylcyclohexane



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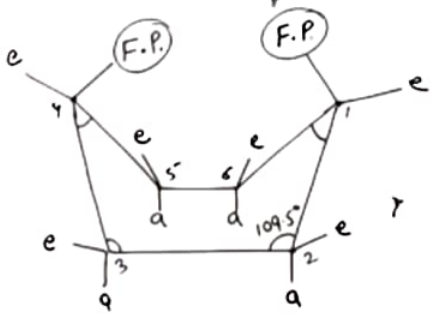
(B) Boat Conformation:-





CONFORMATIONAL ISOMERISM

(B) Boat Conformation:-



=

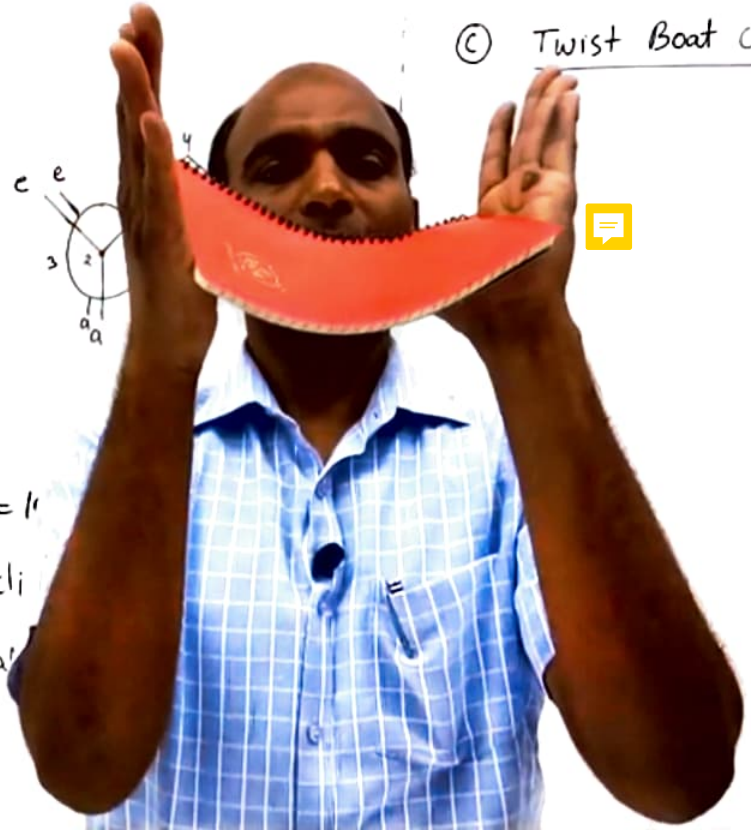
⇒ F.P. = Flag pole position

⇒ No angle Strain (Bond angle = 1

⇒ Max. Torsional Strain (Ecli

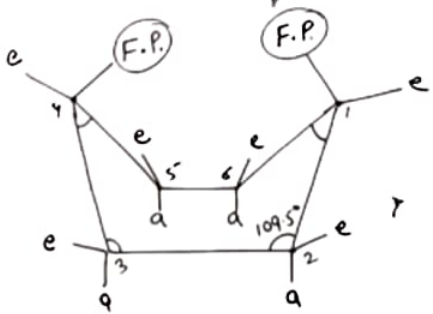
⇒ Repulsion (F.P. – F.P. > axia

© Twist Boat conformation:-



CONFORMATIONAL ISOMERISM

(B) Boat Conformation:-



=

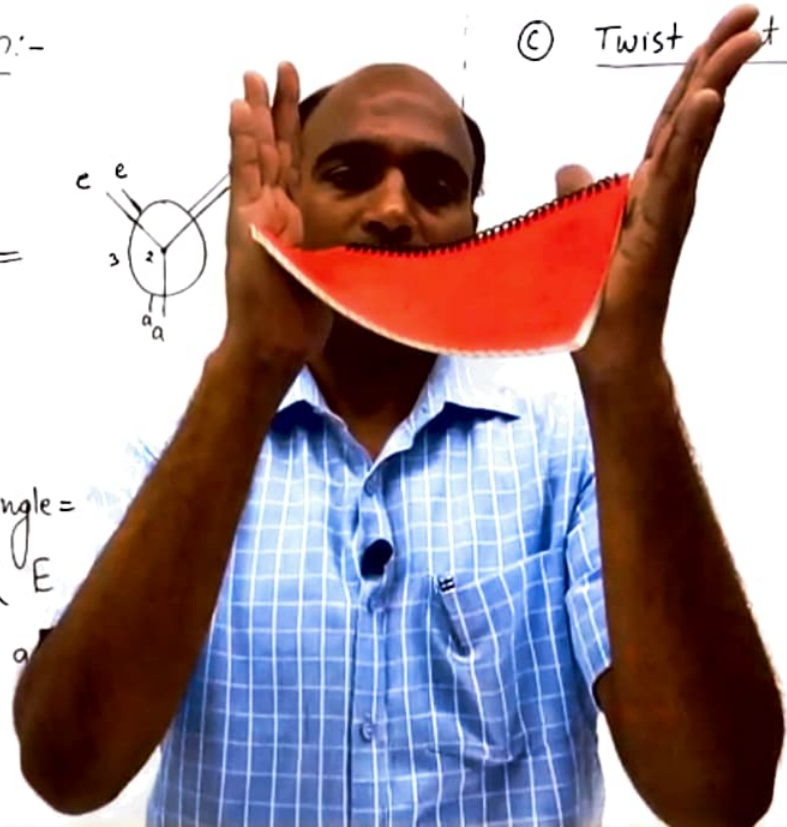
⇒ F.P. = Flag pole position

⇒ No angle Strain (Bond angle =

⇒ Max. Torsional Strain (E

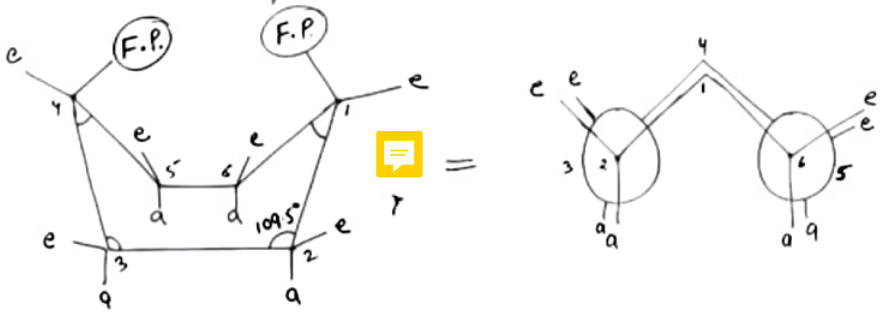
⇒ Repulsion (F.P. – F.P. > a

© Twist t conformation:-



CONFORMATIONAL ISOMERISM

(B) Boat Conformation:-



⇒ F.P. = Flag pole position

⇒ No angle Strain (Bond angle = 109.5°)

⇒ Max. Torsional Strain (Eclipsed conformation)

⇒ Repulsion (F.P. – F.P. > axial - axial)

(C) Twist Boat conformation:-



⇒ Partial angle Strain

⇒ Partial torsional Strain.

(D) Half Chair Conformation:

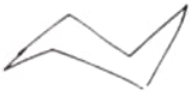


⇒ High angle Strain

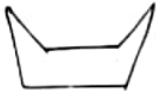
⇒ High Torsional Strain

CONFORMATIONAL ISOMERISM

Plane and center of symm



POS = 3
COS = ✓



POS = 2
COS = X

© Twist Boat conformation:-



POS = X
COS = X
} Chiral

⇒ Partial angle strain
⇒ Partial torsional strain.

(D) Half Chair Conformation

POS = ✓
COS = ✗



- High angle Strain
- High Torsional Strain

CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

⇒ Stability order of conformations of cyclohexane

Chair > Twist Boat > Boat form > Half chair

⇒ Order of interconversion:-

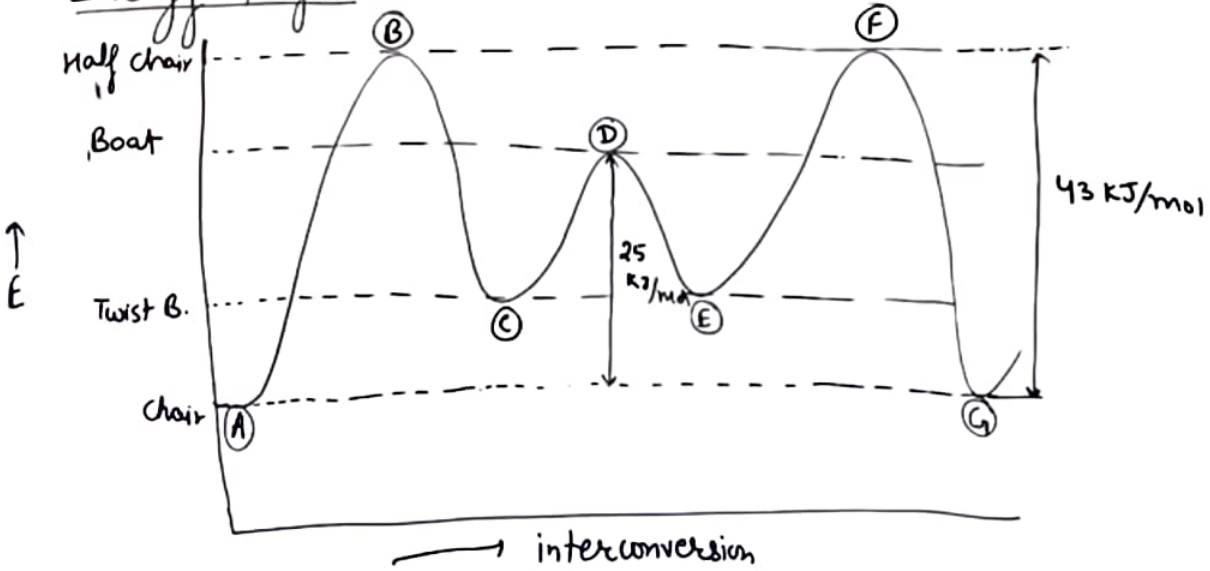
(A) Chair ⇌ (B) Half chair ⇌ (C) Twist Boat ⇌ (D) Boat ⇌ (E) Twist Boat ⇌ (F) Half chair ⇌ (G) Chair



CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

⇒ Energy Profile

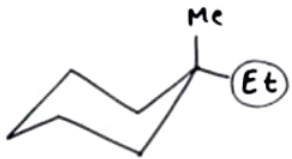


CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

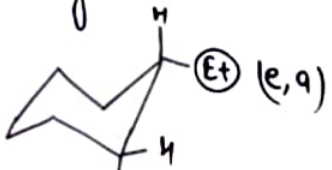
Draw most stable conformation.

① 1-Ethyl-1-methyl cyclohexane →

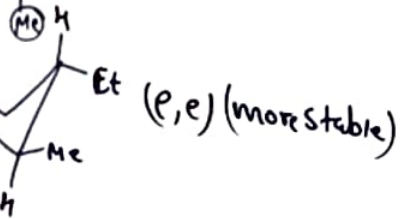


② 1-Ethyl-2-methyl cyclohexane

(2.a) Cis



(2.b) Trans



(e,e) (more stable)

CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

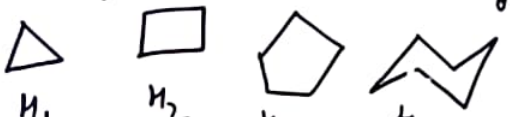
Draw most stable conformation.

Heat of combustion:-

$$C_xH_y + \left(\frac{2x+\frac{y}{2}}{2}\right)O_2 \longrightarrow xCO_2 + \frac{y}{2}H_2O + \text{Heat of comb.}$$

ΔH_{comb}

- Heat of combustion $\propto$ no. of carbon atoms.
- Heat of comb. $\propto \frac{1}{\text{Stability}}$ (for isomer)

eg 

→ Heat of comb. (Per mole) → $H_4 > H_3 > H_2 > H_1$

→ Heat of comb. (Per CH_2) → $\frac{H_4}{6} < \frac{H_3}{5} < \frac{H_2}{4} < \frac{H_1}{3}$

CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

Q Draw most stable conformation.

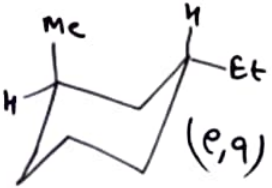
③ 1-Ethyl-3-methyl cyclohexane

(3.a) Cis ⇒ 

(e,e)

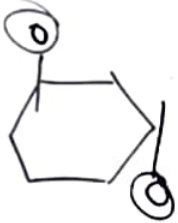
⇒ Stability ⇒ Cis > Trans.

⇒ Heat of comb = Cis < Trans

(3.b) Trans = 

(e,a)





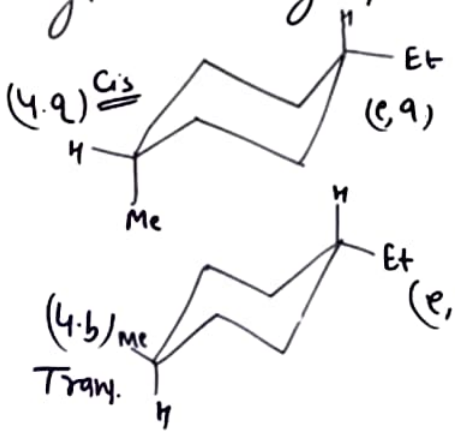


CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

Draw most stable conformation.

④ 1-Ethyl-4-methylcyclohexane



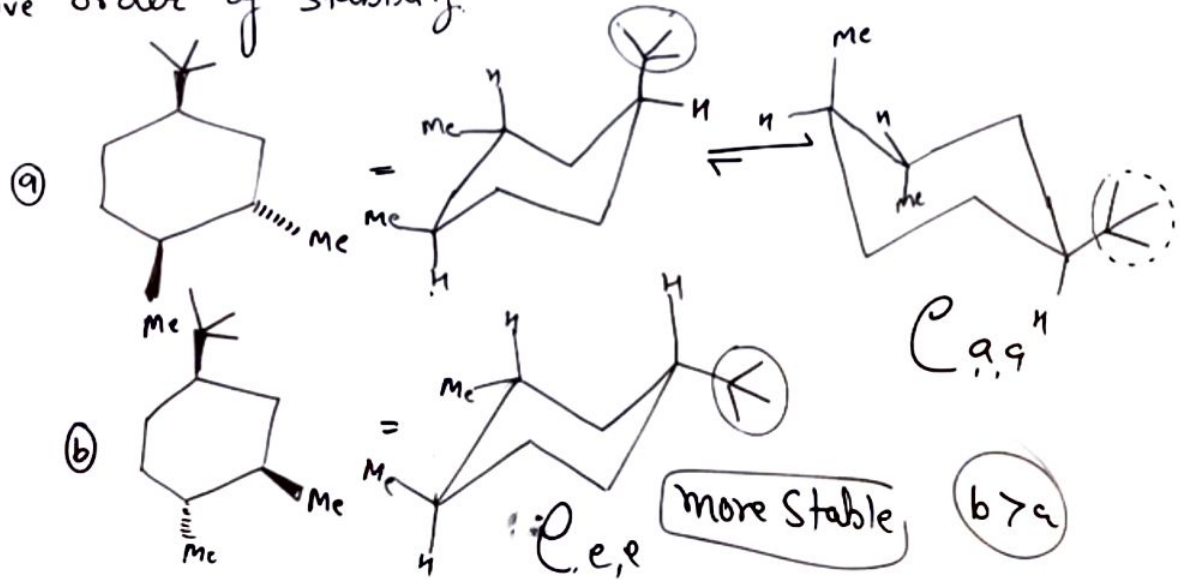
→ Stability = Trans > Cis

→ Heat of comb. = Trans < Cis

CONFORMATIONAL ISOMERISM

CONFORMATIONS OF CYCLOHEXANES

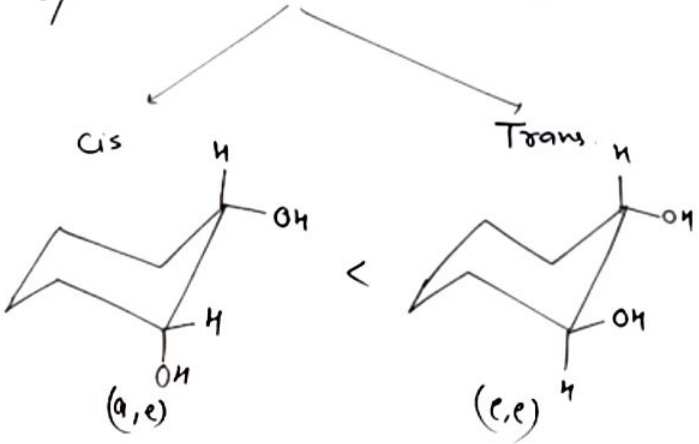
Q Give order of Stability.



CONFORMATIONS OF CYCLOHEXANE

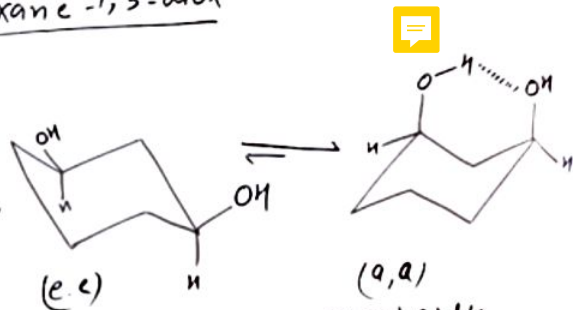
Q Draw most stable conformations

① cyclohexane-1,2-diol ⇌ cis / Trans



② Cyclohexane-1,3-diol

⇒ Cis

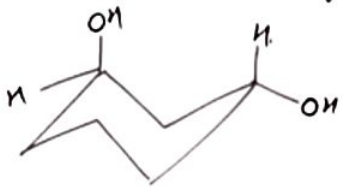


most stable conformation due to intramolecular hydrogen bonding

CONFORMATIONS OF CYCLOHEXANE

Q Draw most stable conformations

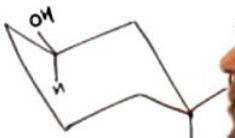
→ Trans



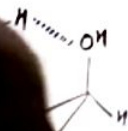
(a,e)

② Cyclohexane-1,3-diol

→ Cis

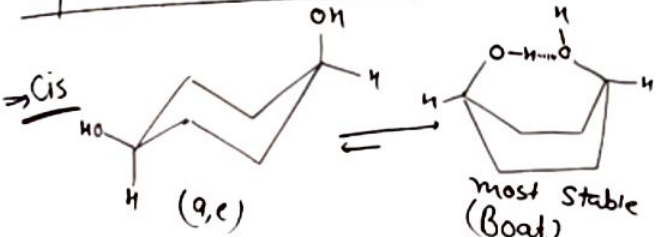


(e,e)



③ Cyclohexane-1,4-diol :-

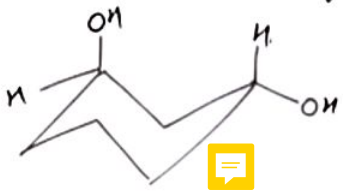
→ Cis



CONFORMATIONS OF CYCLOHEXANE

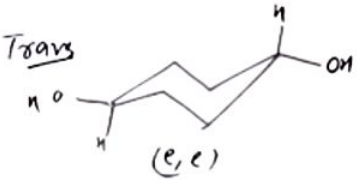
Q Draw most stable conformations

→ Trans



(a,e)

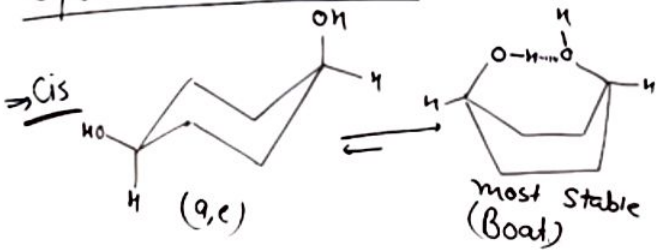
→ Trans



(e,e)

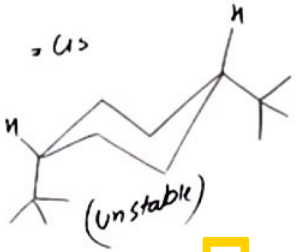
(3) Cyclohexane-1,4-diol :-

→ Cis



(4) 1,4-Di tert butyl cyclohexane

→ Cis



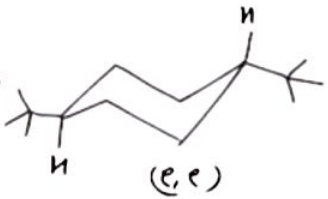
⇌ Twist Boat

most Stable

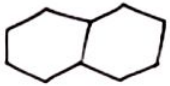
CONFORMATIONS OF CYCLOHEXANE

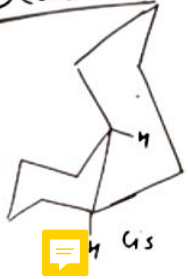
Q Draw most stable conformations

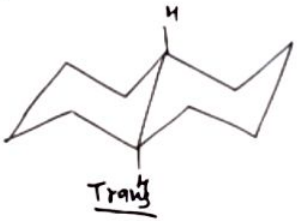
Trans



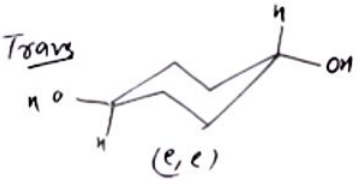
⇒ Decaline



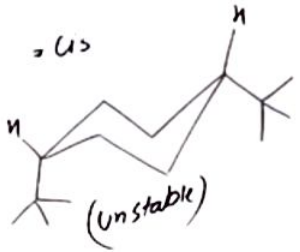




⇒ Trans



④ 1,4-Di tert butyl cyclohexane



⟹ Twist Boat most stable